Simulation of a Physical-Layer Baseband Communication Chain

Introduction and System Overview

This project implements and evaluates, in MATLAB, the complete transmitter–receiver chain of a physical-layer baseband digital communication system. An analog message m(t) was converted into a digital bit stream through sampling, quantization, and source coding (Part I), transmitted over a noisy channel using digital modulation, and recovered through demodulation, detection, and the corresponding decoding stages (Part II). Each block was implemented as a MATLAB local function with the required signature, verified in isolation.

Four modules were carried out in order: (1) signal representation through sampling and sinc reconstruction, including a demonstration of aliasing below the Nyquist rate; (2) two-level and uniform multi-level quantization with mean-square-error (MSE) analysis for M = 4, 16, and 32; (3) source coding of the quantized symbol stream, comparing a fixed-length baseline against an optimal Huffman code built from the empirical symbol distribution and benchmarked against the source entropy; and (4) BPSK and QPSK modulation over an additive white Gaussian noise (AWGN) channel, with bit-error-rate (BER) estimation over 10^6 bits and comparison to the theoretical Q-function curve.

1 Signal Representation through Sampling

1.1 Sampling and reconstruction functions

The sampling stage was implemented as two functions with the required signatures. [t_sample, x_sample] = sample(t, xt, fs) extracts samples of the densely defined waveform at the uniform instants $t(1) + n/f_s$ by interpolating the dense grid, returning the sampled time indices and values. xr = reconstruct(t, x_sample, fs) rebuilds the signal on the original time axis through ideal sinc interpolation, $xr(t) = \Sigma_n\, x[n]\cdot \mathrm{sinc}(f_s(t - t_s[n]))$, where $t_s[n] = t(1) + n/f_s$ which the sampling theorem guarantees to be exact for band-limited signals sampled at $f_s \geq 2f_{max}$.

1.2 Sampling below and above Nyquist

To emulate a continuous-time signal in MATLAB, $x(t) = \cos(2\pi f_0 t)$ with $f_0 = 100$ Hz was generated on a dense time grid (f_dense = 10 kHz), and the function sample() was used to extract samples at two rates relative to the Nyquist rate $f\_N = 2f_0 = 200$ Hz. Figure 1 (left) shows both cases: at $f_s = 0.5f\_N = 100$ Hz, the sinc reconstruction produced by reconstruct() does not recover the original tone but instead locks onto a low-frequency alias, giving a reconstruction MSE of 1.51. At $f_s = 2fn = 400$ Hz, the same interpolation formula recovers the signal almost exactly (MSE = 1.56×10^{-4}), consistent with the sampling theorem's requirement $f_s \geq 2f_{max}$. All reconstruction MSEs were evaluated on the interior of the observation window (10%–90%) to exclude the edge error caused by truncating the sinc sum to a finite record. In the frequency domain (Figure 1, bottom), the properly-sampled reconstruction keeps its line at 100 Hz while the under-sampled one collapses to 0 Hz ,the same aliasing, now seen as a folded spectral peak. Each block's output feeds the next, so an error in sampling would propagate through quantization and coding, which is why each stage was verified in isolation before integration.

This is the sampling theorem in action: a signal is perfectly recoverable from its samples only when the sampling rate exceeds twice its highest frequency, so that no two distinct frequencies produce the same sample sequence. Below that rate the 100 Hz tone and its 0 Hz alias become indistinguishable from the samples alone, and no reconstruction method, however ideal, can separate them. The failure is therefore fundamental to the sampling, not a limitation of the sinc interpolator: information lost at the sampling step cannot be recovered downstream.

1.3 Message signal and reconstruction

For the message, the two-tone waveform $m(t) = 0.7\cdot\cos(2\pi\cdot 250t) + 0.5\cdot\sin(2\pi\cdot 400t)$ was chosen. Dual-tone signaling is an interesting application in its own right where every key of a dual-tone signal in the spirit of DTMF keypad is encoded as exactly such a superposition of two known frequencies, and it makes the frequency content known by construction, so suitable input parameters could be determined directly: $f_{max} = 400$ Hz gives a Nyquist rate of 800 Hz, and the sampling rate was set with a comfortable margin at $f_s = 1600$ Hz. Figure 1 (right) shows m(t), its samples, and the sinc reconstruction lying on top of one another; the reconstruction MSE of $\mathbf{7.71 \times 10^{-6}}$ confirmed that the message is recovered essentially perfectly assuming an ideal remainder of the chain, so any distortion measured in later modules is attributable to quantization and channel noise rather than to sampling. A longer 2-second record of m(t) was then sampled at the same rate (3201 samples) so that the quantizer MSEs and symbol statistics of Modules 2–3 are well averaged.

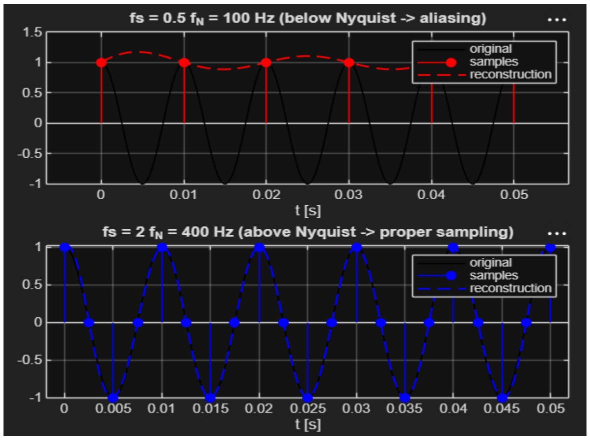


Figure 1 (left): cosine sampled at 0.5f_N (aliasing) and 2f_N (proper sampling).

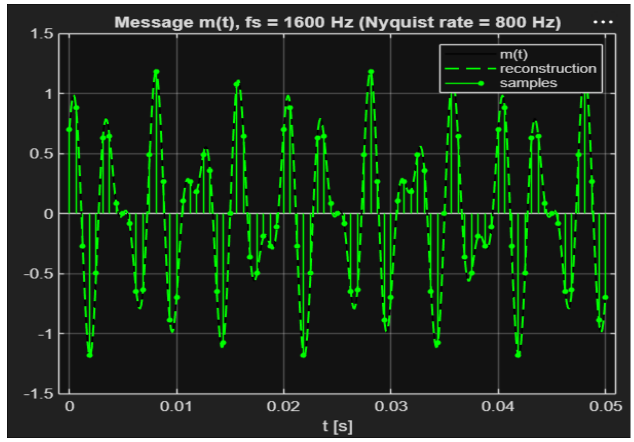


Figure 1 (right): message m(t) at fs = 1600 Hz with samples and sinc reconstruction.

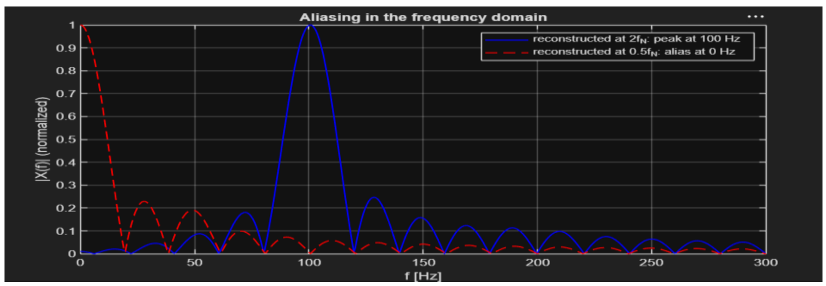
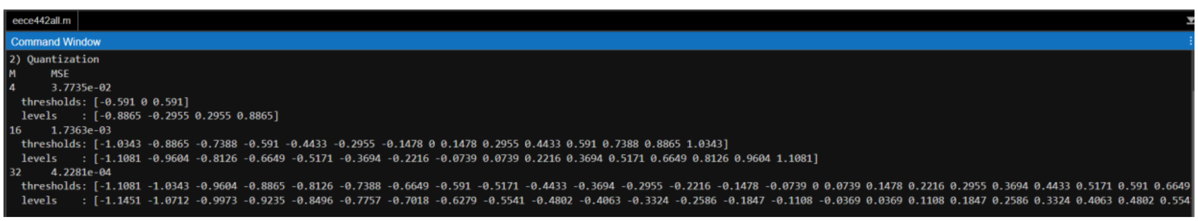


Figure 1 (bottom): aliasing in the frequency domain — the properly-sampled reconstruction (blue) keeps its 100 Hz line, while the under-sampled one (red) collapses to 0 Hz, the folded alias.

2 Quantization



```
eece442all.m
Command Window
2) Quantization
M       MSE
4       3.7735e-02
  thresholds: [-0.591 0 0.591]
  levels    : [-0.8865 -0.2955 0.2955 0.8865]
16      1.7363e-03
  thresholds: [-1.0343 -0.8865 -0.7388 -0.591 -0.4433 -0.2955 -0.1478 0 0.1478 0.2955 0.4433 0.591 0.7388 0.8865 1.0343]
  levels    : [-1.1081 -0.9604 -0.8126 -0.6649 -0.5171 -0.3694 -0.2216 -0.0739 0.0739 0.2216 0.3694 0.5171 0.6649 0.8126 0.9604 1.1081]
32      4.2281e-04
  thresholds: [-1.1081 -1.0343 -0.9604 -0.8865 -0.8126 -0.7388 -0.6649 -0.591 -0.5171 -0.4433 -0.3694 -0.2955 -0.2216 -0.1478 -0.0739 0 0.0739 0.1478 0.2216 0.2955 0.3694 0.4433 0.5171 0.591 0.6649
  levels    : [-1.1451 -1.0712 -0.9973 -0.9235 -0.8496 -0.7757 -0.7018 -0.6279 -0.5541 -0.4802 -0.4063 -0.3324 -0.2586 -0.1847 -0.1108 -0.0369 0.0369 0.1108 0.1847 0.2586 0.3324 0.4063 0.4802 0.554
```

2.1 Quantizer function and two-level quantization

Quantization was implemented as xq = quan(x, thr, levels): the M -1 sorted thresholds partition the amplitude axis into M decision regions (a sample's region index is one plus the number of thresholds strictly below it), and each sample is replaced by the level of its region. Two-level quantization was applied first, using a single mid-range threshold at (x_{min} + x_{max})/2 and the midpoints of the two half-ranges as levels. Figure 2 (left) shows the result: the output collapses to a binary ±0.59 waveform that preserves only which side of mid-range each sample lies on, and the distortion is severe , all amplitude detail is lost, visible as large gaps between x and xq.

2.2 Uniform multi-level quantization

For uniform M-level quantization, the measured dynamic range of the sampled message, [x_{min}, x_{max}] ≈ [−1.182, +1.182], was divided into M equal intervals of width $\Delta = (x_{max} - x_{min})/M$; the M − 1 interval boundaries served as thresholds and the interval midpoints as levels. This was run at low rate (M = 4) and high rate (M = 16 and M = 32). At M = 4 (Figure 2, right) the staircase tracks the envelope only coarsely, with steps of Δ = 0.591 and MSE = 3.773×10^{-2}; at M = 16 and M = 32 (Figure 3 left &right) the quantized waveform progressively hugs the samples until the two curves are nearly indistinguishable at plot scale. The thresholds and levels used in each case, printed in full by the script, are summarized in Table 1. Figure 3 (bottom) overlays both quantizers on the message; the Lloyd-Max levels (red) visibly pull toward the signal's peaks near ±1.1, where the uniform levels (blue) leave larger gaps , the visual cause of the 38% MSE drop.

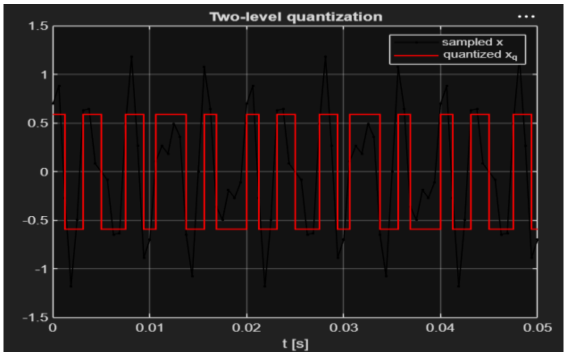


Figure 2 (left): two-level quantization of the sampled message.

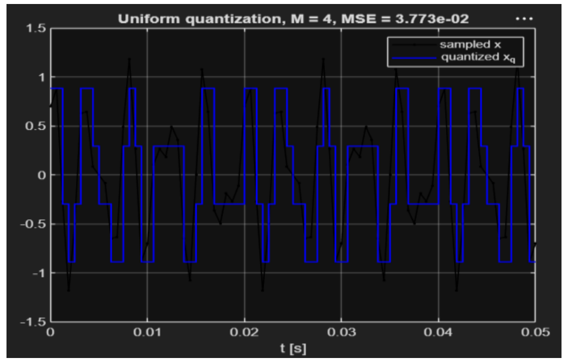


Figure 2 (right): uniform quantization, M = 4, MSE = 3.773×10⁻².

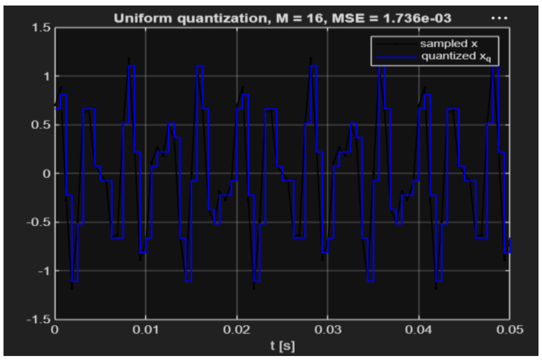


Figure 3 (left): uniform quantization, M = 16, MSE = 1.736×10⁻³.

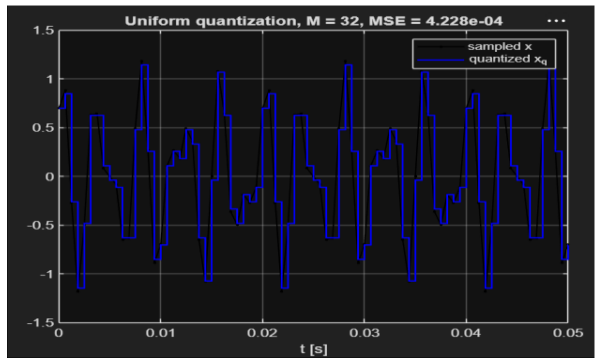


Figure 3 (right): uniform quantization, M = 32, MSE = 4.228×10⁻⁴.

M	Step Δ	Thresholds (uniform grid)	Levels (interval midpoints)	MSE
4	0.591	[−0.591, 0, 0.591]	[−0.8865, −0.2955, 0.2955, 0.8865]	3.7735e−02
16	0.1478	±{0.1478k}, k = 0…7 (15 thresholds)	±{0.0739 + 0.1478k}, k = 0…7	1.7363e−03
32	0.0739	±{0.0739k}, k = 0…15 (31 thresholds)	±{0.0369 + 0.0739k}, k = 0…15	4.2281e−04

Table 1: thresholds, levels, and MSE for each uniform quantization rate.

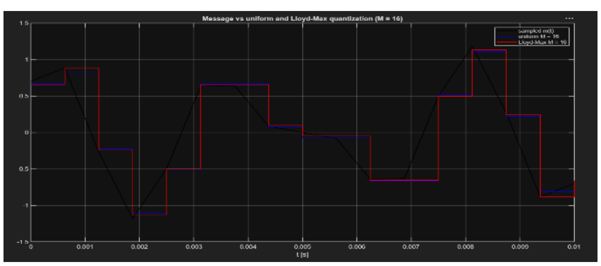
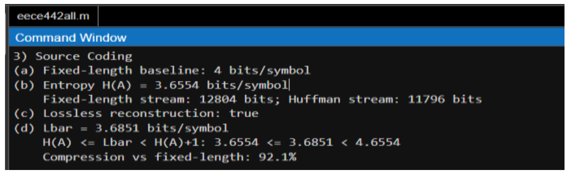


Figure 3 (bottom): message m(t) with its uniform (blue) and Lloyd-Max (red) M = 16 quantizations. The Lloyd-Max levels pull toward the signal's peaks near ±1.1, where the uniform grid leaves larger gaps — the visual cause of the 38% MSE reduction.

Comparing the MSE across rates (Figure 4, left, log scale) confirmed the expected $\Delta^2/12$ behavior of uniform quantization. Quadrupling M from 4 to 16 should reduce the MSE by a factor of 16. The measured factor was 21.7 rather than 16 because the $\Delta^2/12$ law assumes the quantization error is uniformly distributed within each interval a high-rate assumption that no longer holds at M = 4, where only four wide intervals cover the dynamic range and the signal's amplitude density is far from flat inside them. At M = 16 the assumption is already accurate (measured MSE 1.736×10^{-3} vs. $\Delta^2/12 = 1.819\times10^{-3}$). Doubling M from 16 to 32 gave a measured factor of 4.1 against the theoretical 4 essentially exact, i.e., each added bit of quantization rate buys about 6 dB of signal-to-quantization-noise ratio.

Beyond the required uniform quantizer, an MSE-optimal Lloyd-Max quantizer (Topic 3) was trained on the message with the uniform levels as initial guess and applied through the same quan() function; MSE fell from 1.736×10^{-3} to 1.080×10^{-3} at the same rate , a 38% reduction (+2.1 dB SQNR) ,with the optimized levels concentrating where the amplitude density peaks. This is visible in Figure 3 (bottom): near the signal's extremes around ±1.1 the red Lloyd-Max steps sit noticeably closer to the black curve than the blue uniform steps, while both nearly coincide in the mid-range where samples are dense.

3 Source Coding and Decoding



The M = 16 quantized stream was compressed following the four required steps. Although the quantizer defines 16 levels, only 14 symbols actually occur, so the Huffman dictionary was built over those 14 non-zero-probability symbols; the fixed-length baseline is $\lceil \log_2 14 \rceil = 4$ bits/symbol.

(a) With the distribution unknown, the no-model baseline assigns every symbol a fixed-length code of $\lceil \log_2 |A| \rceil$ = **4 bits/symbol**. Encoding the full 3,201-symbol stream with this fixed-length code produced 12,804 bits, against 11,796 bits for the Huffman-encoded stream of Section (c) the counted totals behind the 92.1% ratio reported below.

(b) The empirical distribution was then learned from the stream itself as p(a) = #{a}/N, and the entropy computed as $H(A) = -\Sigma\, p(a)\log_2 p(a)$ = **3.6554 bits/symbol**. The distribution (Figure 4, right) is clearly non-uniform: symbols 4 and 13 each occur with probability ≈ 0.125 because the levels of symbols 4 and 13 sit at ∓0.66, and each tone of the message spends most of its time near its turning points, concentrating probability away from zero, inner symbols cluster near 0.09, outer symbols fall to 0.03, and two of the 16 symbols never occur at all , which is exactly why the entropy sits below the 4-bit baseline.

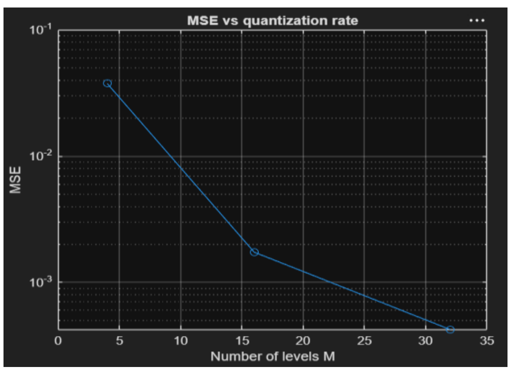


Figure 4 (left): quantizer MSE vs. number of levels M (log scale).

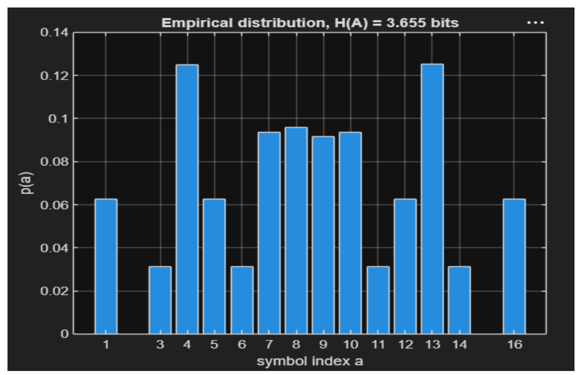


Figure 4 (right): empirical symbol distribution of the M = 16 quantized message, H(A) = 3.655 bits.

(c) A Huffman code was built for source A from the empirical probabilities using huffmandict, the stream was encoded to a single bit sequence and decoded back by huffmanenco / huffmandeco, and the decoded stream was verified to be identical to the input lossless reconstruction confirmed by assertion.

(d) The average code length came out to $\bar{L} = \Sigma\, p(a)\cdot l(a)$ = **3.6851 bits/symbol**, satisfying the source coding theorem bound $H(A) \le \bar{L} < H(A) + 1$ ($3.6554 \le 3.6851 < 4.6554$) with a redundancy of only 0.0297 bits/symbol ,within 0.8% of the entropy limit. Against the fixed-length baseline the compressed stream is 92.1% of the original size, a 7.9% rate saving at zero distortion. The saving is modest because the distribution, while non-uniform, is not heavily skewed; Huffman is optimal among instantaneous codes, yet optimal does not mean zero redundancy. Integrated into the chain, this stage reduces the number of bits handed to the modulator, directly lowering transmission time at no cost in fidelity.

Repeating the full source-coding pipeline on the Lloyd-Max stream gave H = 3.6401 and $\bar{L}$ = 3.6560 bits/symbol (11,703 total bits vs 11,796), so the optimized quantizer improved distortion and rate simultaneously, which is an instance of the Lloyd-Max reduced both MSE and bit count, and the "across quantizers and coding strategies" assessment the module asks for. Concentrating levels where samples are dense makes some symbols more probable and others rarer, sharpening the distribution, which is precisely what lets Huffman compress it further.

4 Modulation, AWGN Channel, and BER Estimation

4.1 Simulation model

Modulation and demodulation were constructed at constellation level with the built-in functions pskmod and pskdemod. BPSK maps each bit to ±1 on the real axis; QPSK carries 2 bits/symbol on a Gray-mapped 4-point constellation with $\pi/4$ phase offset. Following the required procedure, a random bit sequence of length 10^6 was generated (fixed seed for reproducibility), modulated with BPSK first and then QPSK, and passed through the AWGN channel using the built-in awgn function with the 'measured' option at every E_b/N_0 from 0 to 10 dB. Because a QPSK symbol carries two bits ($E_s = 2E_b$), its per-symbol SNR was set to $E_b/N_0 + 10\log_{10}2$ so that both schemes are compared at the same energy per bit. The noisy symbols were demodulated by minimum-distance detection, and each BER was estimated by counting the discrepancies between the transmitted and received bit sequences and dividing by 10^6.

4.2 Results

Eb/N0 (dB)	BPSK (simulated)	QPSK (simulated)	Theory Q(√(2Eb/N0))
0	7.816e-02	7.898e-02	7.865e-02
1	5.605e-02	5.650e-02	5.628e-02
2	3.768e-02	3.759e-02	3.751e-02
3	2.285e-02	2.293e-02	2.288e-02
4	1.246e-02	1.251e-02	1.250e-02
5	5.895e-03	5.940e-03	5.954e-03
6	2.313e-03	2.450e-03	2.388e-03
7	7.910e-04	7.740e-04	7.727e-04
8	1.960e-04	1.880e-04	1.909e-04
9	2.700e-05	3.100e-05	3.363e-05
10	4.000e-06	2.000e-06	3.872e-06

Table 2: simulated BER of BPSK and QPSK over AWGN vs. theory, 10^6 bits per point

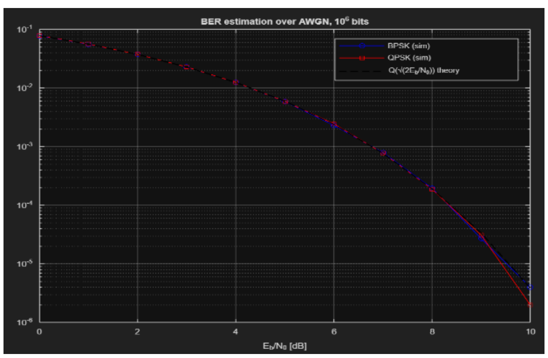


Figure 5: BER of BPSK and QPSK over AWGN (10^6 bits) against the theoretical curve Q(√(2Eb/N0)).

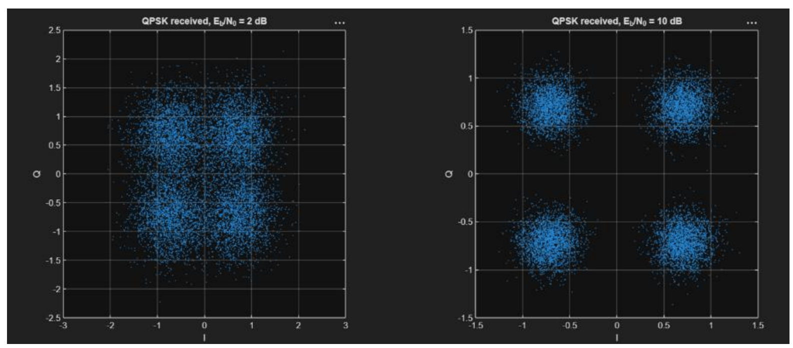


Figure 6: received QPSK constellations at E_b/N_0 = 2 dB and 10 dB (equal axes): the noise clouds around the four Gray-mapped symbols shrink with SNR, which is why the BER in Figure 5 falls.

Both simulated curves in Figure 5 and Table 2 fall on the theoretical waterfall $Q(\sqrt{2E_b/N_0})$ across the whole range, within about 1–3% up to 8 dB. This is the expected , and important comparison result: with Gray mapping and coherent detection, QPSK achieves the same BER as BPSK at equal E_b/N_0, because its two bit streams ride on orthogonal I and Q components that AWGN corrupts independently, so QPSK doubles spectral efficiency at no cost in energy per bit. At 9–10 dB the points scatter visibly (e.g., BPSK 4.0×10^{-6} against a theoretical 3.9×10^{-6}, but QPSK 2.0×10^{-6}): at a BER of $\approx 4 \times 10^{-6}$ only about 4 errors occur in 10^6 bits, so the estimate itself is a random variable with large relative variance. This is a limitation of the Monte-Carlo procedure rather than of the communication link. Reliable estimation at a given BER requires roughly 100/BER transmitted bits to observe about 100 errors.

Conclusion

All required blocks were implemented and validated through the sampling, quantization, source-coding, and BER experiments. Sampling at twice the Nyquist rate reconstructed the two-tone message with MSE $\approx 7.7 \times 10^{-6}$, while sub-Nyquist sampling demonstrated catastrophic aliasing (MSE 1.51); uniform quantization exhibited the predicted 6 dB-per-bit distortion decay from M = 4 (3.77×10^{-2}) to M = 32 (4.23×10^{-4}); Huffman coding of the M = 16 stream reached 3.6851 bits/symbol against an entropy of 3.6554, a lossless 7.9% saving over the fixed-length baseline; and BPSK/QPSK transmission over AWGN reproduced the theoretical Q-function BER over 10 dB of E_b/N_0 using 10^6-bit Monte-Carlo runs. Taken together, the modules quantify the classic design trade-offs of a digital communication system: sampling rate versus bandwidth, quantization rate versus distortion, code length versus entropy, and transmit energy versus error rate.